

FAIR ISLE, United Kingdom

WMO: 030080

Lat: 59.526N

Lon: 1.627W

Elev: 187

StdP: 14.60

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
2	31.2	33.2	21.4	16.1	35.1	23.3	17.7	35.9	47.9	41.1	45.0	40.8	19.9	330	0.886

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	5.2	60.0	57.8	58.5	56.6	57.5	55.7	58.4	59.5	57.2	58.0	56.2	57.0	12.6	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
57.7	71.6	58.8	56.6	68.7	57.7	55.7	66.4	56.6	25.4	59.7	24.6	58.1	24.0	57.2	63.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 41.9	37.0	33.1	DB	29.1	63.4	2.8	2.7	27.1	65.3	25.5	66.9	24.0	68.3	22.0	70.3	(4)
(5)			WB	25.9	59.9	2.8	2.2	23.8	61.5	22.2	62.8	20.6	64.1	18.5	65.7	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	46.8	41.5	40.6	41.4	43.2	46.2	50.1	53.4	54.4	52.6	49.2	45.3	42.9	(6)	
	DBStd	5.78	3.27	3.76	3.66	3.18	2.97	2.40	2.44	2.26	2.50	3.24	3.91	4.09	(7)	
	HDD50	1570	263	264	266	203	120	28	2	1	8	51	144	220	(8)	
	HDD65	6645	728	684	731	653	581	446	359	327	371	489	591	685	(9)	
	CDD50	396	0	0	0	0	3	31	108	138	86	27	3	0	(10)	
	CDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(11)	
	CDH74	0	0	0	0	0	0	0	0	0	0	0	0	0	(12)	
	CDH80	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
	WSAvg	17.1	21.6	21.2	18.8	16.3	13.9	13.1	11.9	12.7	15.7	18.9	20.3	21.4	(14)	
	PrecAvg	39.3	4.7	3.6	3.3	2.1	1.9	2.0	2.2	3.0	3.2	4.3	4.5	4.4	(15)	
(16) Precipitation	PrecMax	48.1	8.5	6.5	6.0	3.2	4.5	4.1	4.7	8.3	5.5	6.2	7.4	7.6	(16)	
	PrecMin	33.1	1.3	1.5	0.7	1.0	0.4	0.6	0.7	1.1	1.1	2.2	2.2	2.0	(17)	
	PrecStd	4.4	1.8	1.3	1.2	0.6	0.8	0.9	1.0	1.4	1.1	1.2	1.3	1.3	(18)	
(20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	48.8	49.2	49.4	50.9	56.3	57.7	63.4	63.1	59.4	55.8	53.5	51.3	(19)	
		MCWB	47.4	47.7	47.0	47.8	53.2	54.4	60.8	59.7	56.6	53.9	51.7	49.9	(20)	
	2%	DB	47.8	48.1	47.9	49.5	52.8	55.7	59.6	60.3	57.7	54.8	52.2	50.3	(21)	
		MCWB	46.4	46.9	46.2	47.2	50.3	53.3	57.9	58.4	55.4	53.1	50.6	49.0	(22)	
	5%	DB	46.9	46.9	46.9	48.3	51.3	54.5	58.2	59.0	56.8	54.0	51.3	49.4	(23)	
		MCWB	45.6	45.5	45.4	46.3	49.0	52.5	56.6	57.0	54.9	52.3	49.8	48.1	(24)	
	10%	DB	46.1	45.8	46.1	47.3	50.2	53.5	56.9	57.9	56.0	53.2	50.4	48.5	(25)	
		MCWB	44.7	44.2	44.6	45.4	48.0	51.7	55.4	56.1	54.3	51.5	48.9	47.1	(26)	
	0.4%	WB	48.3	48.5	47.7	49.0	53.8	55.2	60.4	60.3	57.5	54.9	52.5	50.4	(27)	
		MCDB	48.5	48.7	48.4	49.7	56.0	56.3	62.1	61.4	58.4	55.4	52.9	50.8	(28)	
(30) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	2%	WB	46.8	47.3	46.8	47.6	51.0	53.8	58.2	58.8	56.3	53.9	51.2	49.5	(29)	
		MCDB	47.4	47.8	47.5	48.5	52.2	54.9	59.0	60.2	57.1	54.3	51.8	50.0	(30)	
	5%	WB	46.1	45.9	45.9	46.6	49.6	53.0	56.8	57.6	55.6	53.0	50.2	48.5	(31)	
		MCDB	46.8	46.8	46.7	47.6	50.8	54.0	57.5	58.6	56.3	53.7	51.0	49.2	(32)	
	10%	WB	45.0	44.5	44.9	45.8	48.6	52.1	55.7	56.6	54.8	52.2	49.2	47.3	(33)	
		MCDB	46.1	45.7	45.9	46.8	49.7	53.1	56.4	57.5	55.7	53.0	50.3	48.3	(34)	
	5% DB	MDBR	4.3	4.8	4.8	5.0	5.3	5.0	4.9	5.2	4.5	4.4	4.4	4.8	(35)	
		MCDBR	4.7	5.1	5.1	5.3	6.3	5.8	6.3	6.2	4.9	4.4	4.7	5.1	(36)	
		MCWBR	5.5	5.5	5.3	4.6	5.0	4.7	5.1	4.9	4.3	5.1	5.7	6.4	(37)	
		MCWBR	4.7	5.0	4.9	5.0	5.7	5.4	5.9	5.6	4.4	4.3	4.6	5.1	(38)	
(40) Clear-Sky Solar Irradiance	taub		0.333	0.352	0.367	0.384	0.373	0.369	0.382	0.369	0.363	0.348	0.329	0.438	(40)	
		taud	2.112	2.228	2.296	2.303	2.406	2.455	2.430	2.468	2.449	2.339	2.168	2.536	(41)	
	Ebn at Noon		132	202	241	259	272	274	267	262	243	203	134	95	(42)	
		Edh at Noon	16	23	29	34	33	32	32	29	25	21	15	11	(43)	
(44) All-Sky Solar Radiation	RadAvg		86	268	639	1145	1598	1599	1472	1203	742	369	123	54	(44)	
	RadStd		8	31	43	89	105	142	126	106	58	34	10	5	(45)	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (15 neighbors)	+0.58	N/A	+1.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page